SEMINAR 1

23.02.20..

$$\lim_{n\to\infty}\int_a^b f_n(x)\,dx$$?

1) $$\lim_{n\to\infty}\int_0^1 \underbrace{\frac{x\sin nx}{n^2+nx^2+x}}_{f_n(x)}\,dx = ?$$

Th: $\left.\begin{array}{l} f_n \xrightarrow[{[a,b]}]{cu} f \\ f_n \text{ int}(R) \text{ pe } [a,b] \end{array}\right\} \Rightarrow f \text{ int}(R) \text{ și } \lim_{n\to\infty}\int_a^b f_n(x) = \int_a^b f(x)\,dx$

SOL:

Fie $f_n : [0,1] \to \mathbb{R}$, $f_n(x) = \frac{x\sin nx}{n^2+nx^2+x}$.

$\boxed{CS}$

Fie $x \in [0,1]$ fixat.

$$\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \underbrace{x\sin nx}_{\text{mărg}} \cdot \underbrace{\frac{1}{n^2+nx^2+x}}_{\frac{1}{\infty}}$$

$= 0$

$\Rightarrow f_n \xrightarrow[{[0,1]}]{cs} f \equiv 0.$

$\boxed{CU}$

$$? \lim_{n\to\infty} \underbrace{\sup_{x\in[0,1]} |f_n(x) - f(x)|}$$

$$0 \le \sup_{x\in[0,1]} \left|\frac{1}{n^2+nx^2+x} \cdot x\sin nx\right| \le \frac{1}{n^2}$$

0

$\Rightarrow f_n \xrightarrow[{[0,1]}]{cu} f$

Deci $$\lim_{n\to\infty}\int_0^1 f_n(x)\,dx = \int_0^1 f(x)\,dx = 0$$

2) $\lim_{n\to\infty} \int_0^1 nx(1-x^2)^n \, dx = ?$

Fie $f_n : [0,1] \to \mathbb{R}$, $f_n(x) = nx(1-x^2)^n$

$\boxed{CS}$

Fie $x \in [0,1]$ fixat.

$\lim_{n\to\infty} f_n(x) = 0.$

$\Rightarrow f_n \xrightarrow[{[0,1]}]{CS} f \equiv 0$

$\boxed{CU}$ $\lim_{n\to\infty} \underbrace{\sup_{x\in[0,1]} |f_n(x) - f(x)|}$

$\sup_{x\in[0,1]} nx(1-x^2)^n \geq$

$\underset{x_0 = \frac{1}{n}}{\geq} \left(1 - \frac{1}{n^2}\right)^n \xrightarrow{n} 1 \neq 0$

$\Rightarrow f_n \xrightarrow[{[0,1]}]{\not CU} f \Rightarrow$ nu putem comuta limita

$\lim_{n\to\infty} \underbrace{n \cdot \int_0^1 x(\underbrace{1-x^2}_{t})^n \, dx}_{I}.$ $\quad dt = -2x\,dx.$

$x = 0 \Rightarrow t = 1$

$I = -\frac{n}{2} \cdot \int_1^0 t^n \, dt =$ $\quad x = 1 \Rightarrow t = 0$

$= \frac{n}{2} \cdot \frac{t^{n+1}}{n+1} \Big|_0^1 = \frac{n}{2(n+1)} \xrightarrow{n} \frac{1}{2}$

3) $\lim_{n\to\infty} \int_0^1 \ln(1+x^n) \, dx = ?$

(egalitatea iese din întâmplare)

Integrale improprii

Fie $-\infty < a < b \leq \infty$, $f : [a,b) \to \mathbb{R}$ local integrabilă. Definim $\boxed{\int_a^{b-} f(x)\,dx} = \int_a^b f(x)\,dx$ sn. integrală improprie (generalizată) a fct f pe $[a,b)$.

Dacă $\exists\, L = \lim_{t \to b} \int_a^t f(x)\,dx =$ finită $\Rightarrow$

$\Rightarrow \int_a^b f(x)\,dx$ convergentă

Altfel $\int$ divergentă

- Analog pt $-\infty \leq a < b < \infty$, $f : (a,b] \to \mathbb{R}$
$\int_{a+}^b f(x)\,dx \overset{not}{=} \int_a^b f(x)\,dx$.

- Pt. $-\infty \leq a < b \leq \infty$, $f : (a,b) \to \mathbb{R}$ definim int. improprie $\int_a^b f(x)\,dx$ și spunem că $\int$ conv dacă $\exists\, c \in (a,b)$ a.î. $\int_a^c f(x)\,dx$, $\int_c^b f(x)\,dx$ conv.

1) <u>Natura integralei</u>

$I_1 = \int_0^1 \frac{1}{x^\alpha}\,dx$, $I_2 = \int_1^\infty \frac{1}{x^\alpha}\,dx$ $\alpha \in \mathbb{R}$

$I_3 = \int_0^\infty x \cdot e^{x}\,dx$, $I_4 = \int_1^2 \frac{1}{x^2(x-1)}\,dx$

$I_5 = \int_1^\infty \frac{\cos x}{x^3}\,dx$, $I_6 = \int_2^\infty \frac{\sqrt[3]{x^2+1}}{\sqrt{x^4+x+2}}\,dx$.

$\boxed{I_1, I_2, I_3, I_4 \to \text{definiția}}$

$^* I_7 = \int_0^\infty \frac{\sin x}{x}\,dx = ??$

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Criterii de convergență : I_5, I_6, I_7.

- $I_0 = \int_1^{\infty} x \cdot e^x \, dx$.

$L = \lim_{t \to \infty} \int_1^t x \cdot e^x \, dx$.

$\int_1^t x \cdot e^x \, dx = x \cdot e^x \Big|_1^t - \int_1^t e^x \, dx =$

$= t \cdot e^t - e - e^x \Big|_1^t =$

$= t \cdot e^t - e - (e^t - e) =$

$= t \cdot e^t - \cancel{e} - e^t + \cancel{e} = (t-1) \cdot e^t$

$L = \lim_{t \to \infty} \int_1^t x \cdot e^x \, dx = \lim_{t \to \infty} (t-1) \cdot e^t = \infty$

$\Rightarrow \int_1^{\infty} x \cdot e^x \, dx$ este divergentă

- $I_1 = \int_0^1 \frac{1}{x^\alpha} \, dx$

$\lim_{t \searrow 0} \int_t^1 \frac{1}{x^\alpha} \, dx \overset{\alpha \neq 1}{=} \lim_{t \searrow 0} \frac{x^{-\alpha+1}}{-\alpha+1} \Big|_t^1$

$= \lim_{t \searrow 0} \frac{1}{1-\alpha} (1 - t^{1-\alpha}) =$

$= \begin{cases} \frac{1}{1-\alpha}, & 1-\alpha > 0 \Rightarrow \alpha < 1 \Rightarrow \int \text{conv} \\ \infty, & 1-\alpha < 0 \Rightarrow \alpha > 1 \Rightarrow \int \text{div} \end{cases}$

$\alpha = 1 \Rightarrow \lim_{t \searrow 0} \int_t^1 \frac{1}{x} \, dx =$

$= \lim_{t \searrow 0} \ln x \Big|_t^1 =$

$= \lim_{t \to 0} -\ln t = \infty \Rightarrow \int \text{div}$.

$\Rightarrow I_1 = \begin{cases} \text{conv}; & \alpha < 1 \\ \text{div}; & \alpha \geq 1 \end{cases}$

• $I_2 = \int_1^{\infty} \frac{1}{x^{\alpha}} dx = \lim_{t \to \infty} \int_1^{t} \frac{1}{x^{\alpha}} dx =$

$\alpha \neq 1$

$= \lim_{t \to \infty} \frac{1}{1-\alpha}\left(1 - t^{1-\alpha}\right) = \begin{cases} \frac{1}{1-\alpha}, \alpha > 1 \Rightarrow conv \\ -\infty, \alpha < 1 \Rightarrow div \end{cases}$

$\alpha = 1 \Rightarrow \lim_{t \to \infty} \ln|x| \Big|_1^t =$

$= \lim_{t \to \infty} \ln t = \infty$

$I_2 = \begin{cases} conv, & \alpha > 1. \\ div, & \alpha \leq 1. \end{cases}$

• $I_4 = \int_1^2 \frac{1}{x^2(x-1)} dx$

$\frac{Ax+B}{x^2} + \frac{C}{x-1} = \frac{1}{x^2(x-1)}$

$(x-1)(Ax+B) + Cx^2 = 1$

$\Rightarrow A + C = 0.$

$\left.\begin{array}{l} -A + B = 0 \\ -B = 1 \Rightarrow B = -1 \end{array}\right\} \Rightarrow A = -1 \Rightarrow C = 1.$

$\Rightarrow I_4 = \lim_{t \searrow 1} \int_t^2 \frac{1}{x^2(x-1)} dx =$

$= \lim_{t \searrow 1} \int_t^2 \left(\frac{-x-1}{x^2} + \frac{1}{x-1}\right) dx =$

$= \lim_{t \searrow 1} \int_t^2 \left(-\frac{1}{x} - \frac{1}{x^2} + \frac{1}{x-1}\right) dx =$

$= \lim_{t \searrow 1} \left(-\ln|x| + \frac{1}{x} + \ln|x-1|\right)\Big|_t^2 =$

$= \lim_{t \searrow 1} \Big(-\ln 2 + \frac{1}{2} + \underbrace{\ln t}_{\to 0} - \underbrace{\frac{1}{t}}_{\to 1} - \underbrace{\ln(\underbrace{t-1}_{\to 0})}_{\to -\infty}\Big)$

$= \infty$

$\Rightarrow I_4$ este divergentă.

Ce folosim dacă def nu se poate aplica?

P • $f:[a,b) \to \mathbb{R}$ $\boxed{\text{mărg}}$, local int $\Rightarrow$
$\int_a^b f(x)\,dx$ convergentă

$\longrightarrow \int_0^1 \underbrace{\frac{\sin x}{x}}_{f(x)}\,dx,\ \int_0^1 \frac{tg\,x}{x}\,dx,\ \int_1^2 \frac{\arcsin(x-2)}{x-2}\,dx$

$\lim\limits_{\substack{x\to 0\\ x>0}} \frac{\sin x}{x} = 1. \Rightarrow f$ mărg pe $(0,1]$.

1) <u>Crit. Abel_Dirichlet</u> $\int_a^b f(x)\cdot g(x) = ?$

$f,g:[a,b) \to \mathbb{R}$ și
1) $f \searrow$ și $\lim\limits_{x\to b} f(x) = 0$.
2) $\exists M>0$ a.î. $\left|\int_a^c g(x)\,dx\right| \le M,\ \forall c\in(a,b)$

At $\int_a^b f(x)\cdot g(x)$ conv.

$\longrightarrow \int_1^{\infty} \frac{\sin^2 x}{x^2+x+1}\,dx = \int_1^{\infty} \frac{1}{x^2+x+1} \sin^2 x\,dx$ ($\searrow f$)

$\int_2^{\infty} \frac{\cos x}{x+7}\,dx$

$\int f$ abs conv
dc
$\int |f|$ conv

2) <u>Crit. comparației</u>

$f,g:[a,b) \to [0,\infty)$
$f(x) \le g(x),\ \forall x$
• Dc $\int g$ conv $\Rightarrow \int f$ conv
• Dc $\int f$ div $\Rightarrow \int g$ conv

$\longrightarrow \int \frac{\vdots}{\vdots} < I_1 / I_2$

3) <u>Crit. comp. cu limita</u> $\int_a^b f(x)\,dx = ?$

$g(x) = ?$ a.î. $\lim\limits_{x\to b} \frac{f(x)}{g(x)} = \exists$ finită

$\Rightarrow \int_a^b f(x)\,dx \sim \int_a^b g(x)\,dx$

4) Crit. Cauchy $f:[a,\infty)\to[0,\infty)$

$$\int_a^\infty f(x)\,dx \sim \sum_{n\geq p} f(n)$$

p ↳ primul nr. nat $\geq a$.

- $I_5=\int_1^\infty \frac{\cos x}{x^3}\,dx \to$ ABEL DIRICHLET

$$=\int_1^\infty \underbrace{\frac{1}{x^3}}_{f(x)}\cos x\,dx$$

$f(x)$; $\lim_{x\to\infty} f(x)=0$.

$$\left|\int_1^c \cos x\,dx\right|=\underbrace{|\sin c-\sin 1|\leq 2}_{\int \text{conv}},\ \forall c\in[1,\infty)$$

Crit. comp: $\left|\frac{\cos x}{x^3}\right|\leq \frac{1}{x^3}=g(x)$

$$\int_1^\infty g(x)\,dx=\int_1^\infty \frac{1}{x^3}\,dx \Rightarrow \text{conv} \Rightarrow$$

$\Rightarrow \int |f|$ conv $\Rightarrow$ $\boxed{\int f \text{ abs. conv} \Rightarrow \Rightarrow \int f \text{ conv.}}$

- $I_6=\int_2^\infty \frac{\sqrt[3]{x^2+1}}{\sqrt{x^4+x+2}}\,dx.$

Crit comp cu limită: $g(x)=x^{\frac{2}{3}-2}=x^{-\frac{4}{3}}=\frac{1}{x^{\frac{4}{3}}}$

$\Rightarrow \lim_{x\to\infty}\frac{f(x)}{g(x)}=1$ ($\exists$ limită) $\Rightarrow$

$$\Rightarrow \int_2^\infty f \sim \int_2^\infty \frac{1}{x^{\frac{4}{3}}}\,dx \sim \int_1^\infty \frac{1}{x^{\frac{4}{3}}}\,dx$$

I_2

$\Rightarrow$ convergentă

Cauchy: $\int_2^\infty f(x)\,dx \sim \sum_{n\geq 2} f(n)=$

$$=\sum_{n\geq 2}\frac{\sqrt[3]{n^2+1}}{\sqrt{n^4+n+2}}\sim$$

$$\sim\sum_{n\geq 2}\frac{n^{\frac{2}{3}}}{n^2}=\sum_{n\geq 2}\frac{1}{n^{\frac{4}{3}}}\sim\sum_{n\geq 1}\frac{1}{n^{\frac{4}{3}}}$$ serie arm. div

* • dem. că $\int_0^\infty \frac{\sin x}{x}dx$ conv și calc. valoarea ei.

$$I=\int_0^\infty \frac{\sin x}{x}dx$$

I - convergentă:

$$I=\underbrace{\int_0^1 \frac{\sin x}{x}dx}_{I_1}+\underbrace{\int_1^\infty \frac{\sin x}{x}dx}_{I_2}$$

• $I_1=\int_0^1 \frac{\sin x}{x}dx$

$\lim\limits_{x\to 0}\frac{\sin x}{x}=1 \Rightarrow f$ mărg

$\overset{?}{\Rightarrow} I_1$ - conv.

• $I_2=\int_1^\infty \frac{1}{x}\cdot\sin x\,dx$

$f(x)=\frac{1}{x}$ ↘, $\lim\limits_{x\to\infty}f(x)=0$.

$\left|\int_1^c \sin x\,dx\right|=|\cos 1-\cos c|\le 2$ $\forall\, c\in(1,\infty)$ $\Big\}\overset{A-D}{\Longrightarrow}$

$\Rightarrow I_2$ - conv
I_1 - conv $\Big\}\Rightarrow I$ - conv.

$\boxed{I=?}$

Cunoaștem rel. trig:

$$\frac{1}{2}+\cos x+\cos 2x+\ldots+\cos nx=\frac{\sin(n+\frac{1}{2})x}{2\sin\frac{x}{2}}$$

$\forall\, x\in(0,\pi]$

$$\int_0^\pi\left(\frac{1}{2}+\cos x+\ldots+\cos nx\right)dx=\int_0^\pi\frac{\sin(n+\frac{1}{2})x}{2\sin\frac{x}{2}}$$

$$\frac{1}{2}x+\sin x+\ldots+\frac{\sin nx}{n}\Big|_0^\pi$$

$$\underbrace{\frac{1}{2}\pi+\underbrace{\sin\pi}_{0}}_{\frac{\pi}{2}}+\ldots+\underbrace{\frac{\sin n\pi}{n}}_{0}$$

Deci $\int_0^{\pi} \frac{\sin(n+\frac{1}{2})x}{2\sin\frac{x}{2}}\,dx = \frac{\pi}{2}$. (1)

Fie fct. ajutătoare $f:(0,\pi]\to\mathbb{R}$,

$f(x) = \frac{1}{x} - \frac{1}{2\sin\frac{x}{2}}$.

$\int_0^{\pi} f(x)\,dx$ conv.?

$\lim_{x\searrow 0} f(x) = \lim_{x\searrow 0}\left(\frac{1}{x} - \frac{1}{2\sin\frac{x}{2}}\right) =$

$= \lim_{x\searrow 0} \frac{2\sin\frac{x}{2} - x}{2x\sin\frac{x}{2}} =$

$= \lim_{t\searrow 0} \frac{2\sin t - 2t}{2\cdot 2t\cdot\sin t} = \lim_{t\searrow 0}\frac{\sin t - t}{2t\sin t} \overset{\frac{0}{0}}{=}$

$\overset{\frac{0}{0}}{\underset{l'H}{=}} \lim_{t\searrow 0} \frac{\cos t - 1}{2\sin t + 2t\cos t} \underset{l'H}{=}$

$= \lim_{t\searrow 0} \frac{-\sin t}{2\cos t + 2\cos t - 2t\sin t} = \frac{0}{4} = 0$.

$\Rightarrow f$ - mărg. pe $(0,\pi] \Rightarrow$

$\overset{?}{\Rightarrow} \int_0^{\pi} f(x)\,dx$ convergentă

T. Riemann: Fie $f:[a,b]\to\mathbb{R}$ int.

Atunci: $\lim_{n\to\infty}\int_a^b f(x)\sin nx\,dx =$

$= \lim_{n\to\infty}\int_a^b f(x)\cos nx\,dx = 0$.

$\tilde{f} = \begin{cases} f(x), & x\in(0,\pi] \\ \alpha, & x=0 \end{cases}$ $\quad \int_0^{\pi}\tilde{f} = \int_0^{\pi} f$.

$\lim_{n\to\infty}\int_0^{\pi} f(x)\sin(n+\frac{1}{2})x\,dx = 0. \Rightarrow$

$\Rightarrow \lim_{n\to\infty}\int_0^{\pi}\left(\frac{1}{x} - \frac{1}{2\sin\frac{x}{2}}\right)\cdot\sin(n+\frac{1}{2})x\,dx = 0$

$\Rightarrow \lim_{n\to\infty}\left(\int_0^{\pi}\frac{\sin(n+\frac{1}{2})x}{x}\,dx - \underbrace{\int_0^{\pi}\frac{\sin(n+\frac{1}{2})x}{2\sin\frac{x}{2}}}_{=0.\frac{\pi}{2}}\right)$

$$\Rightarrow \lim_{n\to\infty} \int_0^{\pi} \frac{\sin\left(n+\frac{1}{2}\right)x}{x}\,dx = \frac{\pi}{2}$$

$$\left(n+\frac{1}{2}\right)x = t \Rightarrow dx = \frac{1}{n+\frac{1}{2}}$$

$$x = 0 \Rightarrow t = 0$$

$$x = \pi \Rightarrow t = \left(n+\frac{1}{2}\right)\pi$$

$$\Rightarrow \lim_{n\to\infty} \underbrace{\int_0^{\left(n+\frac{1}{2}\right)\pi} \frac{\sin t}{t}\,dt}_{} = \frac{\pi}{2}$$

$$\int_0^{\pi} \frac{\sin t}{t}\,dt = \frac{\pi}{2} \Rightarrow$$

$$\Rightarrow \int_0^{\infty} \frac{\sin x}{x}\,dx = \frac{\pi}{2}.$$

Integrale Euleriene

Gamma $-\ \Gamma(a) = \int_0^{\infty} x^{a-1} \cdot e^{-x}\,dx$ $\quad -y$

1) $\Gamma(1) = 1$

2) $\Gamma(a) = (a-1)\cdot\Gamma(a-1), \forall a > 1$

$\Gamma(n) = (n-1)!$ 3) $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

Beta. $\beta(a,b) = \int_0^1 x^{a-1} \cdot (1-x)^{b-1}\,dx$

1) $\beta(a,b) = \beta(b,a)$

2) $\beta(a,b) = \frac{\Gamma(a)\cdot\Gamma(b)}{\Gamma(a+b)}$

* 3) $\beta(a,b) = \frac{\pi}{\sin a\pi} = \frac{\pi}{\sin b\pi}$

* 4) $\beta(a,b) = \int_0^{\infty} \frac{x^{a-1}}{(1+x)^{a+b}}\,dx$

• $I = \int_{-\infty}^{-1} \sqrt{(-x-1)^5} \cdot e^{x+1} \, dx$ $-y$

$\boxed{x+1=-y} \Rightarrow x=-y-1 \Rightarrow dx=-dy.$

$x=-\infty \Rightarrow y=\infty$

$x=-1 \Rightarrow y=0.$

$I = \int_{\infty}^{0} -\sqrt{(y+1-1)^5} \cdot e^{-y} \, dy =$

$= \int_{0}^{\infty} \sqrt{y^5} \cdot e^{-y} \, dy = \int_0^{\infty} y^{\frac{5}{2}} \cdot e^{-y} \, dy =$ $a-1 \Rightarrow \boxed{a=\frac{7}{2}}$

$= \Gamma\left(\frac{7}{2}\right) = \frac{5}{2} \cdot \Gamma\left(\frac{5}{2}\right) = \frac{5}{2} \cdot \frac{3}{2} \cdot \Gamma\left(\frac{3}{2}\right) =$

$= \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \Gamma\left(\frac{1}{2}\right) = \frac{15\sqrt{\pi}}{8}.$

• $I = \int_0^a x^2 \sqrt{a^2-x^2} \, dx =$ $a>0$

$= \int_0^a x^2 \cdot a \sqrt{1-\left(\frac{x}{a}\right)^2} \, dx =$

$= a \cdot \int_0^a x^2 \sqrt{1-\left(\frac{x}{a}\right)^2} \, dx$

$\frac{x^2}{a^2} = y \Rightarrow x^2 = a^2 y \Rightarrow x = a\sqrt{y} \Rightarrow$

$\Rightarrow dx = a \cdot \frac{1}{2\sqrt{y}} \, dy.$

$x=0 \Rightarrow y=0$

$x=a \Rightarrow y=1$

$I = a \cdot \int_0^1 a^2 y \sqrt{1-y} \cdot a \cdot \frac{1}{2\sqrt{y}} \, dy =$

$= \frac{a^4}{2} \cdot \int_0^1 \sqrt{y} \cdot \sqrt{1-y} \, dy =$

$= \frac{a^4}{2} \cdot \int_0^1 y^{\frac{1}{2}} \cdot (1-y)^{\frac{1}{2}} \, dy = \frac{a^4}{2} \cdot B\left(\frac{3}{2}, \frac{3}{2}\right)$ $a-1$ $b-1$

$B\left(\frac{3}{2}, \frac{3}{2}\right) = \frac{\Gamma\left(\frac{3}{2}\right) \cdot \Gamma\left(\frac{3}{2}\right)}{\Gamma(3)} = \frac{\left(\Gamma\left(\frac{3}{2}\right)\right)^2}{2!}$

$\Gamma\left(\frac{3}{2}\right) = \frac{1}{2} \cdot \Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2}$ $\Rightarrow$

$\Rightarrow B\left(\frac{3}{2}, \frac{3}{2}\right) = \frac{\left(\frac{\sqrt{\pi}}{2}\right)^2}{2!} = \frac{\pi}{8} \Rightarrow \boxed{I = \frac{a^4 \cdot \pi}{16}}$

$$\int_{-3}^{1} \frac{dx}{\sqrt[6]{(x+3)^5(1-x)}} = ?$$

$$\boxed{t = \frac{x+3}{4}} \Rightarrow dt = \frac{1}{4}dx$$

$x = -3 \Rightarrow t = 0$ $\quad x = 4t-3$

$dx = 4dt$

$x = 1 \Rightarrow t = 1.$

$$\int_0^1 \frac{1}{\sqrt[6]{(4t)^5(\underbrace{1-4t+3}_{4-4t})}} \cdot 4\,dt =$$

$$= \int_0^1 \frac{1}{\sqrt[6]{t^5 \cdot (1-t)}}\,dt =$$

$$= \int_0^1 t^{-\frac{5}{6}} \cdot (1-t)^{-\frac{1}{6}} = \beta\left(\frac{1}{6}, \frac{5}{6}\right) =$$

$a-1$ $\quad b-1$ $\quad \cancel{\Gamma}(1) = 1$

$$= \frac{\pi}{\sin\frac{\pi}{6}} = \frac{\pi}{\frac{1}{2}} = 2\pi$$

__Int. Poisson__ $\quad \int_0^\infty e^{-x^2}dx = \frac{\sqrt{\pi}}{2}$

$$I = \int_0^\infty e^{-x^2}dx$$

$-x^2 = -y \Rightarrow x^2 = y \Rightarrow x = \sqrt{y} \Rightarrow$

$\Rightarrow dx = \frac{1}{2\sqrt{y}}dy$

$x = 0 \Rightarrow y = 0$

$x = \infty \Rightarrow y = \infty$

$$I = \int_0^\infty e^{-y} \cdot \frac{1}{2\sqrt{y}}\,dy$$

$$I = \frac{1}{2}\int_0^\infty y^{-\frac{1}{2}} \cdot e^{-y}\,dy = \frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2}.$$

$a-1$

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

$$\beta\left(\frac{1}{2}, \frac{1}{2}\right) = \frac{\Gamma\left(\frac{1}{2}\right) \cdot \Gamma\left(\frac{1}{2}\right)}{\Gamma(1)}$$

$$\Gamma(1) = \int_0^\infty x^{1-1} \cdot e^{-x}dx = \int_0^\infty e^{-x}dx$$

$$= \lim_{t\to\infty} \int_0^t e^{-x}dx = \lim \ldots$$

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$$\beta\left(\frac{1}{2},\frac{1}{2}\right)=\int_0^1 x^{\frac{1}{2}-1}\cdot(1-x)^{\frac{1}{2}-1}\,dx=$$
$$=\int_0^1 x^{-\frac{1}{2}}\cdot(1-x)^{-\frac{1}{2}}\,dx$$

$$\Rightarrow \beta\left(\frac{1}{2},\frac{1}{2}\right)=\int_{\boxed{0}}^{\boxed{1}}\frac{1}{\sqrt{x(1-x)}}\,dx=$$
$$=\int_0^{\frac{1}{2}}\frac{1}{\sqrt{x}\cdot\sqrt{1-x}}\,dx+\int_{\frac{1}{2}}^{1}\frac{1}{\sqrt{x}\cdot\sqrt{1-x}}\,dx.$$

$$\int\frac{1}{\sqrt{x}\cdot\sqrt{1-x}}\,dx=\int\frac{1}{\sqrt{x-x^2}}\,dx=$$
$$=\int\frac{1}{\sqrt{-(x^2-x)}}\,dx=\int\frac{1}{\sqrt{-\left(\left(x^2-x+\frac{1}{4}\right)-\frac{1}{4}\right)}}\,dx=$$
$$=\int\frac{1}{\sqrt{\frac{1}{4}-\left(x-\frac{1}{2}\right)^2}}=\arcsin\frac{\left(x-\frac{1}{2}\right)}{\frac{1}{2}}=$$
$$=\arcsin(2x-1)+C.$$

$$\beta\left(\frac{1}{2},\frac{1}{2}\right)=\lim_{t\searrow 0}\int_t^{\frac{1}{2}}\frac{1}{\sqrt{x(1-x)}}+\lim_{t\nearrow 1}\int_{\frac{1}{2}}^{t}\frac{1}{\sqrt{x(1-x)}}$$
$$=\lim_{t\searrow 0}\left(\arcsin(2x-1)\Big|_t^{\frac{1}{2}}\right)+\lim_{t\nearrow 1}\left(\arcsin(2x-1)\Big|_{\frac{1}{2}}^{t}\right)$$
$$=\lim_{t\searrow 0}\left(\arcsin 0-\arcsin(2t-1)\right)+$$
$$+\lim_{t\nearrow 1}\left(\arcsin(2t-1)-\arcsin 0\right)=$$
$$=\frac{\pi}{2}+\frac{\pi}{2}=\pi$$

Deci $\pi=\dfrac{\Gamma\left(\frac{1}{2}\right)\cdot\Gamma\left(\frac{1}{2}\right)}{1}\Rightarrow \Gamma\left(\frac{1}{2}\right)=\sqrt{\pi}$

- $I=\int_0^1 x^m\cdot(\ln x)^n\,dx \rightarrow \Gamma$
- $I=\int_0^{\frac{\pi}{2}}\sin^m x\cdot\cos^n x\,dx \rightarrow \beta$

- $I=\int_0^1 x^2\cdot\ln^5 x\,dx.$

$\ln x=t\Rightarrow x=e^t\Rightarrow dx=e^t\,dt$

$$I=\int_{-\infty}^{0} e^{2t}\cdot t^5\cdot e^t\,dt$$

$$I=\int_{-\infty}^{0} e^{3t}\cdot t^5\,dt$$

$3t=-y \Rightarrow t=-\frac{y}{3} \Rightarrow dt=-\frac{1}{3}dy$

$t=-\infty \Rightarrow y=\infty$

$t=0 \Rightarrow y=0.$

$$I=\int_{\infty}^{0} e^{-y}\cdot\left(-\frac{y}{3}\right)^5\cdot\left(-\frac{1}{3}\right)dy=$$

$$=-\frac{1}{3^6}\cdot\int_{0}^{\infty} e^{-y}\cdot y^{5}\,dy=$$

$$=-\frac{1}{3^6}\cdot\Gamma(6)=-\frac{1}{3^6}\cdot 5!$$

• $$I=\int_{0}^{\frac{\pi}{2}} \sin^4 x\cos^{17}x\,dx=$$

$$=\int_{0}^{\frac{\pi}{2}} \sin^3 x\cdot\cos^{16}x\cdot\sin x\cos x\,dx=$$

$$=\int_{0}^{\frac{\pi}{2}} (\sin^2 x)^{\frac{3}{2}}\cdot(\cos^2 x)^{8}\cdot\sin x\cos x\,dx$$

$t=\sin^2 x \Rightarrow dt=2\sin x\cos x\,dx \Rightarrow \sin x\cos x\,dx=\frac{1}{2}dt$

$x=0 \Rightarrow t=0$

$x=1 \Rightarrow t=1$

$$I=\frac{1}{2}\cdot\int_{0}^{1} t^{\frac{3}{2}}\cdot(1-t)^{8}=$$

$a-1 \quad b-1$

$a=\frac{5}{2} \quad b=9$

$$=\frac{1}{2}\cdot B\left(\frac{5}{2},9\right)=$$

$$=\frac{1}{2}\cdot\frac{\Gamma\left(\frac{5}{2}\right)\cdot\Gamma(9)}{\Gamma\left(\frac{5}{2}+9\right)}=$$

$\frac{23}{2}$

$$=\frac{1}{2}\cdot\frac{\cancel{\frac{3}{2}}\cdot\cancel{\frac{1}{2}}\cdot\Gamma\left(\cancel{\frac{1}{2}}\right)\cdot 8!}{\frac{21}{2}\cdot\frac{19}{2}\cdots\cancel{\frac{3}{2}}\cdot\cancel{\frac{1}{2}}\cdot\Gamma\left(\cancel{\frac{1}{2}}\right)}=$$

$$=\frac{8!}{\cancel{2}}\cdot\frac{\cancel{2}\cdot 2\cdots 2^{3}}{3\cdot 5\cdot 7\cdots 21}=\frac{8!\cdot 2^{9}}{21!!}$$

- $I = \int_0^{\infty} e^{ax^2+bx+c}\,dx = -y$??? (Poisson)

EXERCIȚII

1. Să se calculeze următoarele integrale:

1. $\int_0^{\infty} x^6 \cdot e^{-3x}\,dx$.

$3x = y \Rightarrow x = \frac{y}{3} \Rightarrow dx = \frac{1}{3}dy$.

$I = \int_0^{\infty} \left(\frac{y}{3}\right)^6 \cdot e^{-y} \cdot \frac{1}{3}\,dy =$

$= \frac{1}{3^7} \cdot \int_0^{\infty} y^{6} \cdot e^{-y} = \frac{1}{3^7} \cdot \Gamma(7) = \frac{6!}{3^7}$.